

AEROFLEX-ANDOVER, NJ, USA

WMO: 724077

Lat: 41.009N

Lon: 74.736W

Elev: 177

StdP: 99.22

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54779

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.3	(c) -12.6	(d) -22.7	(e) 0.5	(f) -10.8	(g) -19.9	(h) 0.7	(i) -10.3	(j) 7.1	(k) -0.3	(l) 6.2	(m) -0.3	(n) 1.3	(o) 0	(p) 0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.0	23.0	30.3	22.3	28.9	21.5	24.6	29.2	23.7	28.0	23.0	26.9	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.3	26.5	22.4	17.5	25.8	21.8	16.8	25.1	75.1	29.3	71.7	28.2	68.8	27.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.7	5.0	4.6		-19.1	34.3	4.0	1.6	-22.0	35.5	-24.3	36.4	-26.5	37.3	-29.4	38.5
			DB	-19.6	26.2	3.8	0.9	-22.4	26.8	-24.6	27.3	-26.7	27.8	-29.4	28.5
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-2.3	-1.4	3.4	10.3	15.7	20.2	22.8	21.9	18.1	11.9	6.0	1.2
	DBStd	9.79	5.58	5.19	5.29	4.80	4.29	3.35	2.70	2.74	3.54	4.50	4.56	4.78
	HDD10.0	1417	382	321	214	52	4	0	0	0	0	32	136	276
	HDD18.3	3187	639	553	463	244	106	19	2	4	47	206	372	531
	CDD10.0	1671	1	1	9	61	181	305	397	367	243	90	15	2
	CDD18.3	399	0	0	0	3	24	74	140	113	39	5	0	0
	CDH23.3	3298	0	0	1	73	253	624	1159	848	307	32	1	0
	CDH26.7	969	0	0	0	19	67	173	387	244	75	4	0	0
Wind	WSAvg	1.5	1.8	1.9	2.0	2.0	1.5	1.2	1.1	1.0	1.1	1.5	1.6	1.7
Precipitation	PrecAvg	1144	86	67	96	89	90	115	101	110	107	100	85	98
	PrecMax	1707	197	181	220	194	198	249	186	329	288	369	130	200
	PrecMin	686	40	17	21	43	22	24	27	33	2	0	32	29
	PrecStd	242	39	32	47	38	40	59	39	61	79	82	28	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	17.0	22.3	29.2	30.8	32.9	34.0	33.5	31.7	26.4	21.1	17.3
		MCWB	13.3	13.0	14.2	17.1	21.1	23.4	23.8	23.5	22.6	19.4	15.5	14.6
	2%	DB	11.8	12.1	17.9	25.2	28.7	30.4	32.0	31.1	28.8	23.6	18.0	13.6
		MCWB	9.0	7.6	11.3	14.7	19.8	22.1	23.4	23.3	21.4	17.8	14.2	11.3
	5%	DB	7.9	8.6	14.4	22.3	26.3	28.5	30.3	29.1	26.8	21.6	16.0	10.8
		MCWB	5.2	5.0	8.9	14.3	18.3	21.2	22.5	22.1	20.0	17.2	12.2	8.3
	10%	DB	5.5	5.8	11.8	18.9	23.8	27.1	28.8	27.7	24.8	19.4	13.5	8.3
		MCWB	3.1	2.7	7.4	11.8	17.1	20.4	21.9	21.2	19.4	15.8	10.2	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	13.9	15.3	19.5	22.7	24.6	25.7	24.5	21.5	17.5	15.3	
		MCDB	15.5	16.9	19.7	25.2	28.6	30.2	31.4	30.4	28.9	23.9	19.6	16.7
	2%	WB	9.7	8.6	12.5	17.1	21.2	23.4	24.6	24.5	22.9	19.8	15.2	11.9
		MCDB	11.0	11.6	16.7	22.6	26.4	28.4	29.5	28.5	26.1	22.2	17.3	13.2
	5%	WB	5.8	5.2	10.0	15.2	20.0	22.5	23.8	23.5	21.8	18.2	13.1	8.8
		MCDB	7.6	7.8	13.5	20.4	24.2	27.0	28.2	27.2	24.9	20.8	15.3	10.3
	10%	WB	3.4	3.0	7.8	13.2	18.7	21.5	23.0	22.6	20.7	16.2	10.9	5.9
		MCDB	5.2	5.5	11.4	17.7	22.3	25.4	26.9	26.0	23.3	18.8	12.8	7.7
Mean Daily Temperature Range		MDBR	9.4	10.0	10.5	11.7	11.5	11.0	11.0	10.9	11.4	10.6	9.6	8.0
	5% DB	MCDBR	13.1	14.1	14.7	16.5	14.5	12.8	12.5	12.2	13.5	13.2	12.9	10.9
		MCWBR	10.8	10.3	9.1	7.9	6.7	5.9	5.2	5.3	6.6	7.5	8.8	9.2
	5% WB	MCDBR	11.7	12.7	12.7	13.7	11.9	11.2	10.5	10.3	10.7	10.7	10.7	10.2
		MCWBR	10.6	10.0	9.5	8.1	6.2	5.8	5.0	5.0	6.0	7.4	8.9	9.9
Clear-Sky Solar Irradiance	taub	0.299	0.318	0.349	0.370	0.413	0.473	0.474	0.461	0.380	0.354	0.319	0.299	
	taud	2.444	2.388	2.387	2.368	2.273	2.140	2.150	2.176	2.380	2.463	2.518	2.514	
	Ebn at Noon	860	892	900	901	866	811	806	804	856	842	832	833	
	Edh at Noon	78	96	108	118	133	152	149	141	108	89	73	68	
All-Sky Solar Radiation	RadAvg	1.79	2.68	3.62	4.65	5.18	5.64	5.83	5.04	4.21	2.88	2.01	1.50	
	RadStd	0.14	0.28	0.29	0.47	0.55	0.52	0.31	0.33	0.48	0.32	0.19	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+64

Nomenclature: See separate page